

AuthiChain - Vercel Production Deployment Guide

Complete guide to deploying AuthiChain to Vercel production

Prerequisites

✓ Completed Steps:

- [x] Application code is production-ready
- [x] Production build tested successfully (`npm run build`)
- [x] All environment variables configured in `.env`
- [x] Cannabis/strain references cleaned (99.2% reduction)
- [x] TypeScript compilation successful
- [x] Vercel configuration files created

⚠ You Need:

- [] Vercel account (free tier works) - [Sign up](https://vercel.com/signup) (<https://vercel.com/signup>)
 - [] Git repository (GitHub, GitLab, or Bitbucket) OR manual deployment
 - [] Production database URL (current: Railway PostgreSQL)
 - [] Stripe Live API keys (if using payments)
 - [] 15 minutes for deployment
-

Deployment Options

Option A: Git-Based Deployment (Recommended)

Best for: Continuous deployment, team collaboration, automatic updates

1. Push to Git Repository

```
bash
cd /home/ubuntu/authichain_premium/app
git init
git add .
git commit -m "Initial commit - AuthiChain production ready"
git remote add origin your-repo-url
git push -u origin main
```

2. Import to Vercel

- Visit: <https://vercel.com/new>
- Click "Import Project"
- Select your Git provider (GitHub/GitLab/Bitbucket)
- Select the `authichain_premium/app` repository
- Vercel will auto-detect Next.js framework

3. Configure Project Settings

- **Framework Preset:** Next.js

- **Root Directory:** `./` (or `app` if monorepo)
- **Build Command:** `npm run build`
- **Output Directory:** `.next`
- **Install Command:** `npm install`
- **Node Version:** 18.x or higher

4. Skip Initial Deployment

- Click "Deploy" but expect it to fail (environment variables not set yet)
- This is normal - we'll add env vars next

Option B: CLI Deployment

Best for: Quick testing, one-time deployment, no Git setup

1. Install Vercel CLI

```
bash
npm install -g vercel
```

2. Login to Vercel

```
bash
npx vercel login
```

Follow the email verification link

3. Deploy

```
bash
cd /home/ubuntu/authichain_premium/app
npx vercel --prod
```

4. Follow Prompts

- Set up and deploy? → Yes
- Which scope? → Your account
- Link to existing project? → No
- Project name? → `authichain`
- Directory? → `./`
- Override settings? → No



Environment Variables Configuration

Method 1: Via Vercel Dashboard (Recommended)

1. Navigate to Project Settings

- Go to <https://vercel.com/dashboard>
- Select your project: `authichain`
- Click **Settings** → **Environment Variables**

2. Add Variables One by One

Copy from your local `.env` file. For each variable:

Variable Name	Value Source	Environments
DATABASE_URL	From .env	Production, Preview
NEXTAUTH_URL	https://your-project.vercel.app	Production
NEXTAUTH_SECRET	From .env	All
STRIPE_SECRET_KEY	Use LIVE key sk_live_...	Production
STRIPE_WEBHOOK_SECRET	From Stripe Dashboard	Production
NEXT_PUBLIC_STRIPE_PUBLISHABLE_KEY	Use LIVE key pk_live_...	All
All 15 STRIPE_PRICE_ID_*	From Stripe Dashboard	All
NFT_STORAGE_API_KEY	From .env	All
PRIVATE_KEY	From .env	Production
POLYGON_AMOY_RPC_URL	From .env	All
CONTRACT_ADDRESS	From .env	All
POLYGONSCAN_API_KEY	From .env	All
All Cloudflare variables	From .env	All
All feature flags	From .env	All

For each variable:

- Click "Add New"
- Enter variable name
- Enter value (paste from your .env)
- Select environments: ☒ Production, ☒ Preview, ☒ Development
- Click "Save"

1. Critical Variables to Update

These MUST be changed from your local .env :

```

```bash
Update this with your actual Vercel deployment URL
NEXTAUTH_URL="https://authichain-YOUR-USERNAME.vercel.app"

Use Stripe LIVE keys (not test keys)
STRIPE_SECRET_KEY="sk_live_..." # NOT sk_test_
NEXT_PUBLIC_STRIPE_PUBLISHABLE_KEY="pk_live_..." # NOT pk_test_
```

```

Method 2: Via Vercel CLI

```
# Set individual variables
npx vercel env add DATABASE_URL production
# Paste value when prompted

# Or use a script (be careful with special characters)
while IFS= read -r line; do
  if [[ $line =~ ^[A-Z_]+=.[ ] ]; then
    var_name=$(echo $line | cut -d '=' -f 1)
    var_value=$(echo $line | cut -d '=' -f 2-)
    echo "Setting $var_name..."
    echo "$var_value" | npx vercel env add "$var_name" production
  fi
done < .env
```

Method 3: Bulk Import (Vercel CLI)

```
# Create a file with ONLY production-ready values
cp .env .env.production

# Edit .env.production to update:
# - NEXTAUTH_URL to Vercel domain
# - Stripe keys to LIVE versions
# - Any other production-specific values

# Import all at once
npx vercel env pull .env.production
```



Post-Deployment Configuration

1. Update NEXTAUTH_URL

After first deployment, update the environment variable:

1. Note your Vercel deployment URL: `https://authichain-xxxxx.vercel.app`
2. Go to Project Settings → Environment Variables
3. Find `NEXTAUTH_URL`
4. Update value to: `https://authichain-xxxxx.vercel.app`
5. Click “Save”
6. Redeploy: Click “Deployments” → ... → “Redeploy”

2. Configure Stripe Webhook

1. Get Webhook Endpoint URL

`https://your-project.vercel.app/api/webhooks/stripe`

2. Add to Stripe Dashboard

- Visit: `https://dashboard.stripe.com/webhooks`
- Click “Add endpoint”

- URL: `https://your-project.vercel.app/api/webhooks/stripe`

- Select events:

- `checkout.session.completed`
- `customer.subscription.created`
- `customer.subscription.updated`
- `customer.subscription.deleted`
- `invoice.payment_succeeded`
- `invoice.payment_failed`
- Click “Add endpoint”

3. Get Signing Secret

- Click on your new webhook endpoint
- Reveal “Signing secret”: `whsec_...`
- Copy this value

4. Update Vercel Environment Variable

- Go to Vercel → Settings → Environment Variables
- Find or add `STRIPE_WEBHOOK_SECRET`
- Paste the signing secret: `whsec_...`
- Save and redeploy

3. Configure Custom Domain (Optional)

1. Add Domain in Vercel

- Project Settings → Domains
- Enter your domain: `authichain.com`
- Click “Add”

2. Update DNS Records

Follow Vercel’s instructions to add:

- A record or CNAME pointing to Vercel

3. Update Environment Variables

```
bash
```

```
NEXTAUTH_URL="https://authichain.com"
```

4. Update Stripe Webhook

- Update webhook URL to: `https://authichain.com/api/webhooks/stripe`

Verification Testing

After deployment, test these critical features:

1. Basic Pages

-  Homepage: `https://your-project.vercel.app/`
-  Explore: `/explore`
-  Pricing: `/pricing`
-  Mint: `/mint`
-  Dashboard: `/dashboard`

2. API Endpoints

```
# Test NFT list
curl https://your-project.vercel.app/api/nft/list

# Test health check
curl https://your-project.vercel.app/api/usage-check
```

3. Wallet Connection

1. Visit your site
2. Click “Connect Wallet”
3. Connect MetaMask
4. Verify wallet address displays correctly

4. Subscription Flow

1. Go to `/pricing`
2. Click “Get Started” on any plan
3. Verify redirects to Stripe Checkout
4. Complete test purchase (use Stripe test card: `4242 4242 4242 4242`)
5. Verify redirect back to `/dashboard/success`
6. Check subscription status in dashboard

5. NFT Minting

1. Go to `/mint`
2. Upload test image
3. Fill in metadata
4. Mint NFT
5. Verify NFT appears in `/dashboard/nfts`

6. Database Connection

- Check Vercel deployment logs for database connection success
- Verify no connection errors in Function Logs



Troubleshooting

Issue: Build Fails

Error: “Module not found” or “Type error”

Solution:

```
# Verify local build works
cd /home/ubuntu/authichain_premium/app
npm run build

# If local build works, check Vercel build logs
# Usually caused by missing dependencies or environment variables
```

Issue: Environment Variables Not Working

Symptoms: Database connection fails, Stripe not working

Solution:

1. Verify all variables are set in Vercel dashboard
2. Check variable names match exactly (case-sensitive)
3. Ensure selected for correct environment (Production)
4. Redeploy after adding variables
5. Check Function Logs for specific error messages

Issue: Stripe Webhook Failing

Error: "Webhook signature verification failed"

Solution:

1. Verify `STRIPE_WEBHOOK_SECRET` is set correctly in Vercel
2. Ensure webhook URL matches deployment URL
3. Check Stripe Dashboard → Webhooks → Events for error details
4. Redeploy after updating webhook secret

Issue: Database Connection Timeout

Error: "Connection timeout" or "ECONNREFUSED"

Solution:

1. Verify `DATABASE_URL` is correct and accessible
2. Check database provider (Railway) isn't blocking Vercel IPs
3. Increase connection timeout in `DATABASE_URL` :
`?connection_limit=5&connect_timeout=15`
4. Check database service is running

Issue: Next.js API Routes 404

Error: API routes return 404 on Vercel but work locally

Solution:

1. Verify file structure: `app/api/route-name/route.ts`
2. Check `vercel.json` rewrites configuration
3. Ensure correct Next.js version (14.2.x)
4. Check deployment logs for route registration

Issue: Wallet Connection Fails

Error: "Network mismatch" or "Provider not found"

Solution:

1. Verify MetaMask is installed and unlocked
2. Check network matches `POLYGON_AMOY_RPC_URL`
3. Ensure `NEXT_PUBLIC_*` variables are set (these must have prefix)
4. Check browser console for specific error

Check Logs

Vercel Dashboard:

- Deployments → Select deployment → Function Logs
- Real-time logs → Click "Logs" tab

Check Build Output:

```
# View last deployment logs  
npx vercel logs your-deployment-url
```

Deployment Checklist

Pre-Deployment

- [x] ☒ Production build successful
- [x] ☒ Environment variables prepared
- [x] ☒ Database accessible
- [x] ☒ Stripe account configured
- [x] ☒ Code committed to Git (if using Option A)

During Deployment

- [] Project created on Vercel
- [] Initial deployment triggered
- [] All 34 environment variables added
- [] `NEXTAUTH_URL` updated with deployment URL
- [] Stripe webhook configured
- [] Stripe Live keys added

Post-Deployment

- [] Homepage loads correctly
- [] API endpoints responding
- [] Database connection working
- [] Wallet connection functional
- [] Subscription flow tested
- [] NFT minting tested
- [] Stripe webhook verified
- [] Custom domain configured (if applicable)
- [] Monitoring enabled

Important URLs

After deployment, bookmark these:

| Service | URL |
|--------------------|---|
| Live Site | <code>https://authichain-xxxxx.vercel.app</code> |
| Vercel Dashboard | <code>https://vercel.com/dashboard</code> |
| Stripe Dashboard | <code>https://dashboard.stripe.com</code> |
| Database Dashboard | <code>https://railway.app</code> (or your provider) |
| NFT.Storage | <code>https://nft.storage</code> |
| Polygon Scan | <code>https://amoy.polygonscan.com</code> |

Monitoring & Analytics

Vercel Analytics

Enable in dashboard:

- Analytics → Enable
- Monitor page views, performance, Core Web Vitals

Vercel Logs

Real-time function logs:

```
npx vercel logs --follow
```

Error Tracking

Consider integrating:

- Sentry: Error tracking
- LogRocket: Session replay
- Datadog: Application monitoring

Future Deployments

Automatic Deployments (Git)

Once connected to Git:

1. Push changes to `main` branch → Auto-deploys to production
2. Push to other branches → Creates preview deployments
3. Pull requests → Automatic preview URLs

Manual Deployments (CLI)

```
cd /home/ubuntu/authichain_premium/app

# Deploy to production
npx vercel --prod

# Create preview deployment
npx vercel
```

Rollback

If something goes wrong:

1. Go to Vercel Dashboard → Deployments
2. Find previous working deployment
3. Click ... → “Promote to Production”

Support Resources

- **Vercel Docs:** <https://vercel.com/docs>
- **Next.js Docs:** <https://nextjs.org/docs>
- **Stripe Docs:** <https://stripe.com/docs>
- **Vercel Support:** <https://vercel.com/support>

Success Criteria

Your deployment is successful when:

1.  Live URL loads homepage without errors
2.  All navigation links work
3.  Wallet connection successful
4.  Pricing page loads with Stripe checkout working
5.  NFT minting creates NFTs successfully
6.  Dashboard displays user data
7.  No critical errors in Function Logs
8.  Stripe webhooks receiving events
9.  Database queries executing successfully
10.  Mobile-responsive and PWA manifest loads

Deployment Prepared: October 19, 2025

Next.js Version: 14.2.28

Node Version Required: 18.x or higher

Estimated Deployment Time: 15-30 minutes

Ready to Deploy! 🚀

Run the helper script to get started:

```
cd /home/ubuntu/authichain_premium/app  
chmod +x deploy-to-vercel.sh  
./deploy-to-vercel.sh
```

Or proceed directly with deployment using Option A or B above.